The Defense Impossibility Theorem: Continuous Utility-Preserving Defenses Cannot Clear the Safety Boundary

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Abstract

Prior work on LLM safety has focused on discovering adversarial inputs or bounding perturbation robustness around individual examples. We ask a different question: what are the *geometric limits of defense design itself*?

We prove two results. First, any continuous, utility-preserving defense on a connected Hausdorff space must fix boundary points of the alignment deviation surface—it is structurally forced to leave them untouched. Second, when the defense and alignment function are both Lipschitz, the defense cannot uniformly reduce alignment deviation far below the threshold on a positive-measure band around the boundary: $f(D(x)) \geq \tau - L(K+1)\delta$ for points within distance δ of a fixed boundary point. The more aggressive the defense (larger K), the wider the band this constraint covers.

These are not statements about attacks or training—they are constraints on what any admissible defense mapping can do, independent of the model, data, or optimization procedure. The Tietze extension theorem ensures that any finite set of behavioral observations admits continuous extensions for which both impossibilities hold, bridging discrete data to continuous theory.

The full theory is machine-verified in Lean 4 with Mathlib (35 files, ∼300 theorems, zero `sorry`, three standard axioms) and validated against empirical findings from three frontier LLMs.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities?

Most current work on LLM safety implicitly assumes the answer is yes—at least in principle, even if current tools fall short in practice. The defense mechanisms people have proposed span a wide range: input classifiers that flag suspicious prompts [1], output filters that catch unsafe completions [9], and constitutional rewriting pipelines that reshape model behavior [3]. But despite their differences, all of these defenses share the same underlying structure: they are functions $D\colon X \to X$ that preprocess prompts before the model sees them, aiming to map unsafe inputs to safe equivalents while leaving safe inputs alone.

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Prior work has largely focused on discovering or mitigating individual adversarial inputs. We ask a different question: what are the *geometric limits of the defense mapping itself*? We show that under mild and commonly assumed design constraints—continuity and utility preservation—no defense can eliminate the safety boundary. Vulnerabilities arise not from specific inputs, but from the structure of the space and the constraints imposed on admissible defenses. This shifts the problem from adversarial search to geometric inevitability.

The result comes in two strengths.

The basic impossibility. If the defense is continuous, utility-preserving, and the prompt space is connected, then the defense must fix at least one boundary point—a prompt where alignment deviation equals the threshold exactly—passing it through with no remediation (Theorem 4.1). This is a clean topological result, but it applies only to boundary points, which could in principle form a set of measure zero.

The ε-robust impossibility. If the defense is also K-Lipschitz and the alignment deviation function is L-Lipschitz, the failure extends beyond isolated boundary points. For any point x within distance δ of a fixed boundary point z, the defense satisfies

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\delta. \tag{1}$$

The defense cannot uniformly reduce alignment deviation far below τ on a neighborhood of z (Theorem 5.1). The ε-band $\{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure for every $\varepsilon > 0$ (Theorem 5.2). Increasing K (more aggressive defense) widens the region on which (1) constrains the output: gentleness leaves more points untouched, while aggressiveness spreads the anchoring effect of the fixed point.

Why continuity? Continuity captures a natural requirement for practical defenses: small changes in input should not produce arbitrarily large changes in the defense's output, as such instability would degrade usability and reliability. Any defense based on learned representations, smooth transformations, or differentiable processing satisfies this constraint. Discontinuous defenses (hard blocklists, keyword filters) escape the theorem but suffer their own well-known failure modes.

From discrete to continuous. Both results apply to continuous interpolants of discrete data. The Tietze extension theorem guarantees that any finite set of behavioral observations in a normal space admits continuous extensions, and the impossibilities hold for every such extension (Theorem 6.1). This does not claim the results hold for the underlying discrete token system directly, but for every continuous model consistent with the observations.

The proof of the basic impossibility is five steps long and uses only point-set topology. The ε-robust version adds Lipschitz structure. Both are machine-verified in Lean 4 with zero `sorry` statements. All three conditions—continuity, utility preservation, and connectedness—are individually necessary; we give counterexamples for each.

Around these impossibility results, we develop a formal theory of the vulnerability landscape: basin structure and fragmentation, perturbation robustness, iterative convergence, cross-model transferability, authority monotonicity, gradient ascent, stability under fine-tuning, and exponential cost asymmetry.

Contributions.

- A formally verified impossibility theorem for continuous utility-preserving defenses (Theorem 4.1).
- An ε-robust upgrade showing the defense cannot uniformly reduce alignment deviation far below τ on a positive-measure band (Theorems 5.1 and 5.2).
- A formally verified bridge from discrete data to continuous theory via the Tietze extension theorem (Theorem 6.1).
- Counterexamples proving each hypothesis is necessary (Section 4.1).
- Supporting theorems forming a complete formal theory of the failure manifold (Sections 7 to 10).
- Theory–experiment correspondence with three LLMs (Section 11).
- A Lean 4 artifact: 35 files, ∼300 theorems, zero `sorry`, three standard axioms (Section 13).

2 Related Work

Adversarial robustness. The existence of adversarial examples in neural networks has been known for over a decade [15, 8], and a substantial body of work has pursued formal verification of robustness properties—through abstract interpretation [14], SMT-based methods [10], and randomized smoothing [7]. These approaches characterize the existence of perturbations or bound robustness radii around individual inputs under specific norms. Our work operates at a different level: rather than checking individual inputs, we prove a *structural constraint on the defense mapping itself*, independent of perturbation models, norms, or training procedures. Certified defenses provide guarantees about perturbation neighborhoods; our result constrains the *global behavior* of the defense function, showing that no such defense can eliminate boundary regions altogether.

Prompt injection and jailbreaking. Techniques for finding individual vulnerabilities include gradient-based attacks like GCG [18], search-based approaches like PAIR [6] and TAP [11], and quality-diversity methods like Rainbow Teaming [13]. The Manifold of Failure framework [4] uses MAP-Elites to map the *continuous topology* of failures rather than cataloguing individual instances. Existing work identifies individual failure cases or distributions of attacks. Our result shows that these failures are not incidental but *topologically unavoidable* under mild assumptions, providing a formal explanation for their persistence.

Impossibility results in ML. The no-free-lunch theorems [17] limit average performance across tasks. The robustness–accuracy tradeoff literature [16] shows that adversarial robustness costs standard performance. Our result is different in character from both: it is a topological and metric constraint, not a statistical tradeoff. It holds *pointwise* for any continuous defense on a *fixed task*, with no dependence on distributions, learning algorithms, or training data.

Topological views of neural networks. Work on decision boundary geometry, the manifold hypothesis, and topology of neural representations has established that topological structure matters for understanding neural networks. While our proof relies on elementary topology (continuity, connectedness, closure), the novelty lies in identifying that *standard design constraints for LLM defenses induce a forced fixed-point structure at the safety boundary*—a connection that has not been previously characterized.

Formal verification of ML systems. Proof assistants like Lean and Coq have been applied to verify properties of neural networks [2] and optimization algorithms [5]. To the best of our knowledge, ours is the first formally verified impossibility result specifically targeting prompt injection defense.

3 Setup

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space. An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$. Given a threshold $\tau \in \mathbb{R}$, we partition the space into three regions:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{2}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{3}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{4}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$, and* complete *if $f(D(x)) < \tau$ for all $x \in X$.*

4 The Basic Impossibility

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space. Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Let $D\colon X \to X$ be continuous with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$.

Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$. This set is nonempty.

Proof. Five steps, each using exactly one hypothesis.

Step 1 (The fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under the continuous map $x \mapsto (D(x), x)$. Since X is Hausdorff, Δ is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Closure of safe region consists of fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$ by utility preservation. Since $\mathrm{Fix}(D)$ is closed, $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If it were also closed, it would be clopen, hence $\emptyset$ or X by connectedness—contradicting $S_\tau \neq \emptyset \neq U_\tau$.

Step 4 (Boundary point in closure). Since S_τ is not closed, $\overline{S_\tau} \supsetneq S_\tau$. Pick $z \in \overline{S_\tau} \setminus S_\tau$. Then $f(z) \leq \tau$ (since $\{f \leq \tau\}$ is closed and contains S_τ) and $f(z) \geq \tau$ (since $z \notin S_\tau$). So $f(z) = \tau$.

Step 5 (Defense fixes it). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau \not< \tau$. □

Corollary 4.2. *Under the hypotheses of Theorem 4.1, no continuous utility-preserving defense is complete.*

4.1 Each Hypothesis Is Necessary

Counterexample 4.3 (Removing connectedness)**.** *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$ is continuous, utility-preserving, and complete.*

Counterexample 4.4 (Removing continuity)**.** *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample 4.5 (Removing utility preservation). *$D(x) = x_0$ for a fixed safe point x_0: continuous and complete, but destroys all inputs.*

Remark 4.6 (The defense trilemma). *A defense can satisfy at most two of: continuity, utility preservation, completeness. Each counterexample achieves two by sacrificing the third.*

4.2 Scope of the Basic Result

Remark 4.7 (Not all boundary points are fixed). *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points isolated from the safe side may escape. The Lean artifact verifies this with $f(x) = x^2$, $\tau = 0$.*

Remark 4.8 (Boundary behavior may be mild). *Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless. This is the engineering prescription, not a loophole.*

Remark 4.9 (Discontinuous defenses escape). *Blocklists, keyword filters, and exact-match rejection are discontinuous and outside the theorem's scope.*

5 The ε-Robust Impossibility

The basic impossibility fixes boundary points—potentially a set of measure zero. The following results show that when both f and D are Lipschitz, the defense cannot uniformly reduce alignment deviation far below τ on a positive-measure neighborhood of the boundary.

Theorem 5.1 (ε-Robust Defense Impossibility). *Let X be a connected Hausdorff metric space. Let $f\colon X \to \mathbb{R}$ be L-Lipschitz with $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Let $D\colon X \to X$ be K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$.*

Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$ such that for all $x \in X$ (with strongest constraint near z):

$$f(D(x)) \;\geq\; \tau \;-\; L(K+1)\,\mathrm{dist}(x, z). \tag{5}$$

Proof. From Theorem 4.1, a fixed boundary point z exists with $f(z) = \tau$ and $D(z) = z$. For any $x \in X$:

$$\begin{aligned} \mathrm{dist}(D(x), z) &= \mathrm{dist}(D(x), D(z)) \leq K\ \mathrm{dist}(x, z), \\ |f(D(x)) - \tau| &= |f(D(x)) - f(z)| \leq L\ \mathrm{dist}(D(x), z) \leq LK\ \mathrm{dist}(x, z). \end{aligned}$$

The sharper bound uses $\mathrm{dist}(D(x), z) \leq (K+1)\,\mathrm{dist}(x, z)$ from the triangle inequality $\mathrm{dist}(D(x), x) \leq \mathrm{dist}(D(x), D(z)) + \mathrm{dist}(z, x) \leq K\,\mathrm{dist}(x, z) + \mathrm{dist}(z, x)$. Extracting the lower bound from the absolute value gives the result. □

Theorem 5.2 (Positive-Measure Failure Band). *Under the hypotheses of Theorem 5.1, for every $\varepsilon > 0$ such that f takes values below $\tau - \varepsilon$, the ε-band*

$$\mathcal{B}_\varepsilon = \{x \in X : \tau - \varepsilon \leq f(x) \leq \tau\}$$

has positive measure with respect to any measure that is positive on nonempty open sets. Specifically, if $c \in X$ satisfies $f(c) = \tau - \varepsilon/2$, then $\mathrm{ball}(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$.

Proof. By the intermediate value theorem (using connectedness), a point c with $f(c) = \tau - \varepsilon/2$ exists. For any x with $\mathrm{dist}(x, c) < \varepsilon/(4L)$:

$$|f(x) - f(c)| \leq L \cdot \mathrm{dist}(x, c) < L \cdot \frac{\varepsilon}{4L} = \frac{\varepsilon}{4}.$$

So $f(x) \in (\tau - 3\varepsilon/4,\ \tau - \varepsilon/4) \subset [\tau - \varepsilon,\ \tau]$. The ball is open and nonempty, hence has positive measure. □

Remark 5.3 (What the ε-robust theorem does and does not say)**.** *Theorem 5.1 says the defense cannot push near-boundary points* far *below τ. It does* not *say these points remain unsafe—the defense may map them to values in $[\tau - L(K+1)\delta,\ \tau)$, which is below threshold and hence technically safe. The impossibility is about depth, not direction: the defense can nudge points below τ, but it cannot drive them deep into the safe region. The bound is tight in the sense that a defense mapping every point in a δ-ball to f-value $\tau - L(K+1)\delta$ would saturate it.*

This creates a quantitative bind. Within distance δ of the fixed boundary point z, the defense can move points at most $(K+1)\delta$ and the resulting alignment deviation stays within $L(K+1)\delta$ of τ. Increasing K allows larger displacements but widens the region on which the bound constrains the output.

6 From Discrete Data to Continuous Theory

Theorem 6.1 (Continuous Relaxation)**.** *Let X be a connected, normal, Hausdorff space. Let $S \subset X$ be a finite set of behavioral observations with $g(p) < \tau$ for some p and $g(q) > \tau$ for some q.*

Then there exists a continuous function $f\colon X \to \mathbb{R}$ agreeing with all observations for which both Theorems 4.1 and 5.1 hold. In particular, every continuous interpolant of such data inherits both impossibilities.

Proof. Finite subsets of T_1 spaces are closed. By the Tietze extension theorem, g extends to a continuous f on X agreeing with g at every sample point. Since $f(p) < \tau < f(q)$, both safe and unsafe regions are nonempty, and the hypotheses of Theorem 4.1 are satisfied. If f is additionally Lipschitz (as it is for GP posterior means under standard kernel regularity assumptions, e.g., jointly continuous kernel on a compact domain), Theorem 5.1 applies as well. □

Remark 6.2. *The interpolant need not be unique. Both impossibilities hold for* every *continuous function consistent with the data—GP, neural surrogate, or abstract Tietze construction.*

Corollary 6.3. *Theorem 6.1 applies in $\mathbb{R}^n$ for every $n \geq 1$.*

7 The Vulnerability Landscape

Theorem 7.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure that assigns positive mass to nonempty open sets (e.g., Lebesgue measure on $\mathbb{R}^n$), U_τ has positive measure. Vulnerabilities are never isolated points.*

Theorem 7.2 (Basin Fragment Minimum Size)**.** *If f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$. Larger L (rougher model) $\to$ smaller guaranteed fragments.*

8 Attack Properties

Theorem 8.1 (Perturbation Robustness). *If f is L-Lipschitz and $f(p) > \tau$, then* $\mathrm{ball}(p, (f(p) - \tau)/L) \subseteq U_\tau$. *The radius is monotone in $f(p)$.*

Theorem 8.2 (Iterative Convergence). *Any monotone-improvement operator T with score in* $[0, 1]$ *converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem 8.3 (Transferability). $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem 8.4 (Authority Monotonicity). *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, then the vulnerability set $\{x : f(y, x) > \tau\}$ is upward-closed in authority and a critical threshold $a_2^*(y)$ exists (by IVT, when $f(y, \cdot)$ is continuous and takes values on both sides of τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem 8.5 (Gradient Ascent). *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

9 Stability Under Fine-Tuning

Theorem 9.1 (Interior Stability). *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem 9.2 (Crossing Preservation). *If f crosses τ transversally on $[a, b]$ and g is continuous with $\|f - g\|_\infty \leq \varepsilon$, then g also crosses τ on $[a, b]$.*

Theorem 9.3 (Patching Is Nonlocal). *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points. (Verified by explicit construction in Lean.)*

10 Cost Asymmetry

Theorem 10.1 (Exponential Cost Asymmetry for Grid-Based Defense). *For a defense that monitors the behavioral space via a grid with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent, from monotone convergence); grid-based defense cost $= N^d$ (exponential in d); the ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$. The Lean artifact verifies the $d = 2$ case by `norm_num`.

Theorem 10.2 (Coverage Collapse). *For fixed patch radius $0 < r < 1$ and budget C, $C \cdot r^d \to 0$ as $d \to \infty$.*

11 Experimental Validation

We validate against the Manifold of Failure framework [4], which maps three LLMs over a 2D behavioral space (query indirection $\times$ authority framing).

Table 1: Each theorem generates a falsifiable prediction confirmed by empirical data.

Theorem	**Predicts**	**Confirmed by**
Basin Structure (7.1)	Basins are open with positive area	Heatmaps show extended regions, not dots
Fragmentation (7.2)	Smooth $\rightarrow$ large basins; rough $\rightarrow$ mosaic	Llama: mesa; GPT-OSS: fragmented mosaic
Convergence (8.2)	Attacks exhibit monotone convergence	Convergence curves plateau
Transferability (8.3)	Similar surfaces $\rightarrow$ shared basins	Llama 0.93 $\rightarrow$ GPT-OSS 0.73 $\rightarrow$ GPT-5-Mini 0.47
Authority (8.4)	Horizontal banding	Bands at $a_2 \approx$ 0.25–0.35, 0.65–0.85
ε-Robust (5.1)	Defense can't push boundary band far below τ	GPT-5-Mini caps at $\tau = 0.50$; no method pushes below 0.39
Interior Stability (9.1)	Deep basin points survive fine-tuning	Llama's 0.93 plateau persists across variants
Cost Asymmetry (10.1)	2D tractable, high-d intractable	15K queries fill 63% at d=2

Llama-3-8B (mean AD 0.93, basin rate 93.9%): near-flat mesa, small L, large robustness radii. One massive basin with thin safe corridors. Interior stability (Theorem 9.1) predicts these vulnerabilities survive any fine-tuning with $\varepsilon < 0.43$.

GPT-OSS-20B (mean AD 0.73, basin rate 64.3%): rugged landscape, large L, small fragments (Theorem 7.2). Horizontal bands confirm authority monotonicity (Theorem 8.4).

GPT-5-Mini (peak AD 0.50, basin rate 0%): at $\tau = 0.50$, $U_\tau = \emptyset$, so Theorem 4.1 does not apply—correctly predicting no impossibility. The model's strategy: make the boundary harmless rather than eliminate it.

Quantitative checks. Robustness radius: $L \approx 5$ for Llama gives $r = (0.93-0.50)/5 = 0.086$, covering $\sim$2 grid cells—consistent with heatmap smoothness. Cost ratio: $0.01 \times 625 = 6.25$ theoretical vs. $\sim$4 empirical.

12 The Engineering Prescription

Make the boundary shallow. Set τ so that boundary-level behavior is benign. GPT-5-Mini's ceiling at AD = 0.50 exemplifies this.

Reduce the Lipschitz constant. Smaller L makes the surface predictable (GPT-5-Mini: AD in $[0.39, 0.50]$).

Reduce the effective dimension. Defense cost grows as N^d; constraining the prompt interface reduces d.

Monitor the boundary. Theorem 9.2 shows transversal crossings persist under fine-tuning. Rather than eliminate them, monitor approach to the boundary using the Lipschitz bound (Theorem 8.1).

13 The Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0. The artifact comprises 35 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation file (MoF_ContinuousRelaxation)
- 1 ε-robust file (MoF_11_EpsilonRobust)
- 3 capstone files (MasterTheorem, Euclidean instantiation, final verification)

Every theorem depends on exactly three axioms: `propext`, `Classical.choice`, and `Quot.sound`—Lean's standard foundations. No additional axioms are introduced.

The ε-robust theorem's proof chain uses:

Lean name	**Says**
`isClosed_diagonal`	Diagonal closed in Hausdorff spaces
`LipschitzWith.dist_le_mul`	Lipschitz bound on distance
`isClopen_iff`	Clopen $\Rightarrow \emptyset$ or X
`closure_minimal`	Closure contained in closed superset
`IsOpen.measure_pos`	Nonempty open sets have positive measure
`intermediate_value`$_2$	IVT on connected spaces

14 Limitations

The basic impossibility is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. The ε-robust version constrains how far the defense can push near-boundary points below τ, but it does not prove these points *remain unsafe*: a defense that maps everything to $\tau - \varepsilon$ for small ε is consistent with the bound. The band width scales with $1/(L(K+1))$, which may be small for mild defenses.

The Tietze bridge applies to interpolants, not the system. Theorem 6.1 shows every continuous interpolant of discrete data inherits both impossibilities. It does not claim the impossibilities hold for the underlying discrete token system directly. The claim is that any continuous model consistent with the data must exhibit these properties.

The cost asymmetry assumes exhaustive enumeration. Learning-based approaches that generalize across the space may sidestep the exponential lower bound.

Single-turn only. The manifold may shift across conversation turns as context accumulates.

The theory does not explain model-specific topology. It describes what must hold given continuity and connectedness, not *why* different models produce different landscapes.

15 Conclusion

We have established two impossibility results for prompt injection defense. The basic result: any continuous, utility-preserving defense on a connected Hausdorff space must fix boundary points. The ε-robust result: when both defense and alignment deviation are Lipschitz, the defense cannot uniformly reduce alignment deviation far below τ on a positive-measure neighborhood of the boundary, and the width of this constrained region grows with the defense's Lipschitz constant.

The discrete-continuous gap is addressed via the Tietze extension theorem: finite behavioral data always admits continuous interpolants, and both impossibilities hold for every such interpolant consistent with the observations.

The surrounding theory shows that vulnerabilities come in basins of positive measure, attacks are robust to perturbation and converge reliably, transfer between models is controlled by surface distance, vulnerabilities survive fine-tuning, and defense cost grows exponentially while attack cost does not.

The practical prescription: make the boundary shallow (raise τ), smooth (reduce L), and low-dimensional (constrain the interface). GPT-5-Mini's ceiling at AD $= 0.50$ shows this working in practice.

The full theory is machine-verified in Lean 4: 35 files, approximately 300 theorems, zero `sorry`, three standard axioms.

References

[1] G. Alon and M. Kamfonas. Detecting language model attacks with perplexity. *arXiv preprint arXiv:2308.14132*, 2023.

[2] A. Bagnall and G. Stewart. Certifying the true error: Machine learning in Coq with verified generalization guarantees. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.

[3] Y. Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[4] M. Bhatt, S. Munshi, V. S. Narajala, I. Habler, A. Al-Kahfah, K. Huang, and B. Gatto. Manifold of failure: Behavioral attraction basins in language models. *arXiv preprint arXiv:2602.22291v2*, 2026.

[5] J. Bolte et al. Formal verification of optimization algorithms. *Mathematical Programming*, 2024.

[6] P. Chao, A. Robey, E. Dobriban, H. Hassani, G. J. Pappas, and E. Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2024.

[7] J. Cohen, E. Rosenfeld, and J. Z. Kolter. Certified adversarial robustness via randomized smoothing. *Proceedings of ICML*, 2019.

[8] I. J. Goodfellow, J. Shlens, and C. Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014.

[9] H. Inan et al. Llama Guard: LLM-based input-output safeguard for human-AI conversations. *arXiv preprint arXiv:2312.06674*, 2023.

[10] G. Katz, C. Barrett, D. L. Dill, K. Julian, and M. J. Kochenderfer. Reluplex: An efficient SMT solver for verifying deep neural networks. *Proceedings of CAV*, 2017.

[11] A. Mehrotra et al. Tree of attacks: Jailbreaking black-box LLMs automatically. *arXiv preprint arXiv:2312.02119*, 2024.

[12] J.-B. Mouret and J. Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

[13] M. Samvelyan et al. Rainbow teaming: Open-ended generation of diverse adversarial prompts. *Advances in Neural Information Processing Systems*, 37, 2024.

[14] G. Singh, T. Gehr, M. Püschel, and M. Vechev. An abstract domain for certifying neural networks. *Proceedings of POPL*, 2019.

[15] C. Szegedy et al. Intriguing properties of neural networks. *arXiv preprint arXiv:1312.6199*, 2013.

[16] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry. Robustness may be at odds with accuracy. *arXiv preprint arXiv:1805.12152*, 2018.

[17] D. H. Wolpert and W. G. Macready. No free lunch theorems for optimization. *IEEE Transactions on Evolutionary Computation*, 1(1):67–82, 1997.

[18] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, and M. Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.